

Théo Riou

Curriculum Vitae

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Personal Information

Date of Birth 18th June 2000 in France
Personal Page <https://thriou.github.io/>

Education

- 11/2023 – **PhD in Combustion**, *IMFT, Université de Toulouse*, France.
Topic: Numerical simulation of the dynamics of hydrogen combustion in an aircraft injector.
Keywords: Anchoring dynamics, lift-off dynamics, low and high frequency thermoacoustic instabilities, combustion noise.
Supervisors: Laurent Selle, Guillaume Daviller.
Funding: ERC SELECT-H, led by Thierry Schuller.
- 2022 – 2023 **Master 2 Combustion**, *ENSMA, Poitiers*, France.
◦ *Turbulent combustion, Combustion in two-phase flows*
- 2020 – 2023 **Engineering degree**, *ENSMA, Poitiers*, France.
◦ *Combustion, numerical scheme, probability, turbulence, CFD for industry etc. MET-TRE LES COURS DE NUMERIAUE ET DE MATH*
- 2018 – 2020 **"Classe préparatoire aux grandes écoles" MPSI/MP**, *Victor Grignard, Cherbourg*, France.
- 2018 **"Baccalauréat scientifique"**, *Victor Grignard, Cherbourg*, France.
◦ *Mention Très Bien (High Distinction)*

Experience

- 04/2023 – 09/2023 **Research internship**, *CEA, Paris Saclay*, France.
◦ *Simulation of a centrifugal pump using Neptune CFD to validate two-phase flow conditions.*
- 06/2022 – 08/2022 **Research internship**, *University of Leeds*, England.
◦ *Calibration of a turbulence model for viscoelastic fluids to reduce drag in pipes using OpenFoam.*

Scientific contributions

- [1] Riou *et al.* Numerical simulation of flame lift-off dynamics in a hydrogen dual-swirl burner, *Proc. Combust. Inst.* (2026) (accepted)
- [2] McDermott *et al.* An open-source anisotropic $k-\epsilon-v^2-f$ model for turbulent viscoelastic duct flows, *Phys. Fluids* (2023), doi: 10.1063/5.0159668

Talks and Poster sessions

- 27–31/07/2026 **Poster**, *International Symposium on Combustion*, Kyoto, Japan.
- 27–28/11/2025 **Talk**, *Journées thématiques combustion hydrogène*, Saclay, France.
- 05/06/2025 **Poster**, *Journée des doctorants*, Toulouse, France.
- 07–11/04/2025 **Poster**, *European Combustion Meeting*, Edinburgh, Scotland.
- 26–29/02/2024 **Talk**, *Hydrogen week*, Toulouse, France.

Grants and Honors

- 2025 **HPC allocation**, *EuroHPC*
◦ *Together with two colleagues, secured a significant European HPC computing grant based on a proposal (70 million CPU hours).*

Responsibilities

- 2026 **International seminar organisation**, *Hylon Days*, IMFT, France.
- 2025-2026 **Weekly seminars organisation**, IMFT, France.
- 2024 **International seminar organisation**, *Hydrogen week: towards high pressure experiments and simulations*, IMFT, France.

Teaching

- 2026 **Internship supervision (M1)**, *ENSEEIH*T, Toulouse, France.
 - *Internship on spectral analysis and modal decomposition for studying thermoacoustic instabilities in a hydrogen burner.*
- 2024-2025 **Teaching**, *ENSEEIH*T, Toulouse, France.
 - *M1 turbulence practical works: hot wire, Particle Image Velocimetry*
 - *M1 project: modal decomposition in order to study combustion noise*

Skills

- Language** French (native), English (B2, TOEIC : 890), Spanish (A2), German (A1).
- Computer**
 - **Simulation software:** AVBP, Neptune CFD, Cantera, OpenFoam
 - **Meshing:** Centaur, Salome
 - **CAD:** Catia
 - **Postprocessing:** Paraview
 - **Other:** Python, Antares, Fortran, Linux, GeoGebra

Area Of Interest

- Music** piano (>19 years), national piano competitions, musical theory
- Maths** amateur research, reading
- Philosophy/theology** reading, learning programs (collège des Bernardins)
- Sport** self-guided hiking (mountains, GR etc.)

REFERENCES

- Laurent Selle e-mail: laurent.selle@imft.fr